

Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch

Abstract

This report presents the implementation of the **Beta-Geometric/Negative Binomial Distribution (BG/NBD)** model from first principles using **NumPy** for numerical computation and **SciPy's L-BFGS-B optimizer** for parameter estimation. The model is used to predict customer purchasing behavior and estimate Customer Lifetime Value (CLV) without relying on specialized marketing analytics libraries.

Introduction

Customer Lifetime Value (CLV) is a key business metric that estimates the total revenue a customer is expected to generate throughout their relationship with a company. This project develops a BG/NBD model from scratch to provide an efficient, transparent, and mathematically grounded approach for CLV estimation.

Theory

The BG/NBD model is based on the following assumptions:

- Customer purchases follow a **Poisson process**.
- Individual transaction rates are **Gamma distributed**.
- Customer dropout probabilities follow a **Beta distribution**.
- Customer behavior exhibits heterogeneity across the population.

Objectives

- Implement the BG/NBD model from scratch using NumPy.
- Estimate the parameters **r , α (alpha), a , and b** using Maximum Likelihood Estimation (MLE).
- Predict future customer transactions.
- Estimate Customer Lifetime Value (CLV).

Methodology

1. Generate or prepare customer transaction summaries as **$(\mathbf{x}, \mathbf{t}_x, T)$** .
2. Implement a vectorized negative log-likelihood function.
3. Optimize model parameters using **SciPy's L-BFGS-B** algorithm.
4. Compute probability alive, expected future transactions, and CLV projections.

Dataset

Synthetic customer transaction data is generated to validate the implementation and verify parameter recovery. The same workflow can be directly applied to real customer transaction datasets after converting purchase histories into $(\mathbf{x}, \mathbf{t}_x, \mathbf{T})$ summary statistics.

Likelihood

The BG/NBD parameters are estimated by maximizing the log-likelihood function through **Maximum Likelihood Estimation (MLE)**. The optimization process minimizes the negative log-likelihood across all customers using the L-BFGS-B optimization algorithm.

Complexity Analysis

The implementation uses fully vectorized NumPy operations, resulting in $O(N)$ time complexity for each likelihood evaluation, where N is the number of customers. Vectorization significantly improves computational efficiency by eliminating explicit Python loops.

Results

The model successfully estimates the parameters $\mathbf{r}$, α , $\mathbf{a}$, and $\mathbf{b}$, calculates customer probability of being alive, predicts expected future transactions over a specified horizon, and projects Customer Lifetime Value (CLV).

Conclusion

This project demonstrates a numerically stable and computationally efficient implementation of the BG/NBD model from first principles. The approach provides a strong foundation for customer analytics research and can be extended to real-world marketing datasets for customer retention and revenue forecasting.

References

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